

Assessment	Test	Domain	Skill	Difficulty
SAT	Reading and Writing	Craft and Structure	Cross-text Connections	Hard

Question

Text 1

What factors influence the abundance of species in a given ecological community? Some theorists have argued that historical diversity is a major driver of how diverse an ecological community eventually becomes: differences in community diversity across otherwise similar habitats, in this view, are strongly affected by the number of species living in those habitats at earlier times.

Text 2

In 2010, a group of researchers including biologist Carla Cáceres created artificial pools in a New York forest. They stocked some pools with a diverse mix of zooplankton species and others with a single zooplankton species and allowed the pool communities to develop naturally thereafter. Over the course of four years, Cáceres and colleagues periodically measured the species diversity of the pools, finding—contrary to their expectations—that by the end of the study there was little to no difference in the pools’ species diversity.

Based on the texts, how would Cáceres and colleagues (Text 2) most likely describe the view of the theorists presented in Text 1?

Answer

- A. It is largely correct, but it requires a minor refinement in light of the research team’s results.
- B. It is not compelling as a theory regardless of any experimental data collected by the research team.
- C. It may seem plausible, but it is not supported by the research team’s findings.
- D. It probably holds true only in conditions like those in the research team’s study.

Correct Answer: C

Rationale

Choice C is the best answer. This is how Cáceres and co. would most likely describe the view presented in Text 1. The view in Text 1 is that historical diversity affects how diverse an ecological community eventually becomes. But Cáceres and co. did not get this result: they found no difference in eventual diversity between a zooplankton pool that started out diverse and a zooplankton pool that started out with only a single species.

Choice A is incorrect. Cáceres and co. would probably not describe the view presented in Text 1 this way. The view in Text 1 is that historical diversity affects how diverse an ecological community eventually becomes. Cáceres and co’s findings directly undermine this view: they found no difference in eventual diversity between a zooplankton pool that started out diverse and a zooplankton pool that started out with only a single species. Choice B is incorrect. Cáceres and co. would probably not describe the view presented in Text 1 this way. Their experiment was designed to test this hypothesis, and their findings were "contrary to their expectations." In other words, before the study, they predicted the theory was correct. Choice D is incorrect. Cáceres and co. would not describe the view presented in Text 1 this way. Their research finding directly undermines the view presented in Text 1: so it definitely doesn’t hold true in conditions like those in the study.